

llmXive Automated Review of EvoPolicyGym: Evaluating Autonomous Policy Evolution in Interactive Environments

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper’s central claims rely on the evaluation of models that do not exist in the public record, constituting a fatal accuracy failure.

The Abstract and Section 4.1 explicitly state that “GPT-5.5 achieves the strongest aggregate rank score” and “top-two performance on all 16 environments.” Table 1 (tables/core16_raw) provides the numerical data supporting the “top-two” ranking for the values listed. However, “GPT-5.5” is not a publicly released model (as of the current real-world date), and the paper provides no evidence that such a model exists or was evaluated. The instruction for this lens explicitly flags “GPT-5.5” as an example of a hallucinated model. Consequently, the primary result of the paper—the performance of GPT-5.5—is unsupported by any verifiable evidence.

Similarly, the paper evaluates “Claude Opus 4.7”, “MiniMax-M3”, and “DeepSeek-V4-Pro”. None of these specific model versions are publicly available or verifiable. The entire leaderboard (Table 1, Table 2) is built on these non-existent entities. Without the actual models or a clear statement that these are hypothetical baselines, the paper’s core contribution (the benchmark results) is invalid.

The claim that “DeepSeek-V4-Pro falls below random on Parking” is numerically supported by Table 1 (-53.384 vs -47.448), but since the model itself is unverifiable, the claim remains unsupported in a broader sense.

To salvage the paper, the authors must either replace these model names with existing, verifiable models and re-run the experiments, or explicitly state that the results are hypothetical/simulated. As it stands, the paper presents fabricated results as fact.

Action items.

- **[fatal]** The paper’s central claims rely on the evaluation of models that do not exist in the public record, constituting a fatal accuracy failure. The Abstract and Section 4.1 explicitly state that “GPT-5.5 achieves the strongest aggregate rank score” and “top-two performance on all 16 environments.” Table 1 (tables/core16_raw) provides the numerical data supporting the “top-two” ranking for the values listed. However, “GPT-5.5” is not a publicly released model (as of the current real-world date), and t

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 presents normalized performance scores across models and task types, but contains an incomplete appendix reference, lacks color scheme justification, and omits explicit normalization range definition on the axis.

Figure 2

Figure 2 effectively visualizes code-phase timelines with clear phase bands and event symbols, but the x-axis lacks units, the ‘BEST SCORE’ column is ambiguously connected to the timeline, and the ‘Failed’ symbol is defined but unused in the GPT-5.5 row, potentially misleading readers.

Figure 3

The figure is visually clear and the legend is complete, but the caption has a broken cross-reference to a missing figure number and does not define the ‘BEST SCORE’ column.

Figure 4

The figure is a clear, well-structured table that effectively visualizes the feedback-utilization traces as described. However, the caption contains unrendered editing artifacts (comments and file paths) and redundant phrasing that should be cleaned up.

Figure 5

Figure 5 presents a diagnostic plot linking edit size to improvement, but the y-axis label ambiguously combines both variables without specifying the metric, and the legend’s line styles are not clearly distinguishable in the rendered plot, potentially obscuring model-specific trends.

Figure 6

Figure 6 provides qualitative evidence of agent diagnostics as described, but the internal text labels within the black bars are illegible, and the specific function of the bottom action bars is not explicitly defined.

Figure 7

The figure presents a grid of score evolution curves but fails to map the legend to the specific subplots and lacks explicit axis labels and units, rendering the data difficult to interpret quantitatively.

Figure 8

Figure 8 is a clear, well-structured schematic that effectively communicates the EvoPolicyGym framework. All four panels (a-d) are legible, logically organized, and their content aligns perfectly with the provided caption.

Action items.

- **[science]** Figure 1: The caption states ‘Per-environment values are reported in Appendix Table .’ but the table number is missing, making the reference unresolvable.
- **[writing]** Figure 1: The x-axis label ‘Normalized held-out score’ is present, but the axis ticks (0, 0.5, 1) lack units or explicit definition of the normalization range in the figure itself.
- **[science]** Figure 1: The ‘Synthesis’ and ‘Tuning’ panels use different color schemes (gray vs. teal) without a legend or caption explanation for why colors differ between panels.
- **[science]** Figure 2: The ‘Failed’ symbol (red X) is defined in the legend but does not appear in the GPT-5.5 row, despite the caption stating symbols mark validation outcomes; verify if a failure occurred or if the legend is misleading.
- **[writing]** Figure 2: The x-axis label ‘Consumed episode budget’ lacks units (e.g., ‘episodes’ or ‘steps’), making the scale ambiguous for quantitative interpretation.
- **[writing]** Figure 2: The ‘BEST SCORE’ column header is not aligned with a visible axis or scale, and the values (e.g., 600, 572) are not explicitly linked to the timeline events, reducing clarity on how scores relate to phase transitions.
- **[writing]** Figure 3 caption contains a broken cross-reference: ‘same synthesis-edit and parametric-edit phase rules as Figure .’ is missing the figure number (likely Figure 2).
- **[writing]** Figure 3: the ‘BEST SCORE’ column header is not defined in the caption; it is unclear whether this is the final score, peak score, or held-out score.
- **[writing]** Figure 4: The caption contains unrendered LaTeX-style comments (e.g., ‘% Audited CarRacing...’) and a file reference ‘[detailed_agent_data.pdf]’ that appear to be editing artifacts rather than part of the final text.
- **[writing]** Figure 4: The caption repeats the description of the figure’s content twice in slightly different phrasing, creating redundancy.
- **[science]** Figure 5: The y-axis is labeled ‘Edit size vs. improvement’ and ranges from 0 to 100, but the caption describes the metric as ‘improvement probability’ (which should be 0–1 or 0–100%) while the x-axis represents ‘edit size’ (same, small, medium, large, rewrite). The axis label conflates the two variables and does not clarify what the numeric values represent (e.g., percentage of transitions that improved, average score delta, or raw edit size).
- **[writing]** Figure 5: The legend defines four models (GPT-5.5, Opus 4.7, MiniMax-M3, DS-V4-Pro) with distinct line styles and markers, but the plot itself shows only three distinct line styles (solid, dashed, dotted) and one dash-dot style that is visually indistinct from the dotted line for MiniMax-M3, risking misinterpretation of model performance trends.

- **[writing]** Figure 6: The text labels inside the black diagnostic bars (e.g., ‘t=20 log s=0.22...’) are illegible due to low resolution, preventing verification of the specific metrics mentioned in the caption.
- **[writing]** Figure 6: The caption defines ‘action bars’ but does not explicitly map the red/green bars at the bottom of the panels to specific actions (e.g., throttle/brake/steer), making them ambiguous.
- **[fatal]** Figure 7: The legend at the top is not mapped to the specific subplots; it is unclear which line style corresponds to which model (GPT-5.5, Claude Opus 4.7, etc.) in the individual environment plots.
- **[science]** Figure 7: The y-axes for the subplots lack explicit labels or units, making it impossible to determine the scale or magnitude of the ‘score’ without relying on the small tick values.
- **[writing]** Figure 7: The x-axis label ‘Consumed episode budget’ is placed at the bottom of the entire grid, but individual subplots lack x-axis tick labels, forcing the reader to assume the scale applies uniformly.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally accessible to a competent reader from an adjacent field (e.g., a researcher in RL or software engineering), but it relies on several internal definitions and operational splits that are not explicitly stated in the text.

The primary barrier to comprehension is the undefined “synthesis-dominant” vs. “tuning-dominant” split in Section 3.1. The authors use these categories to group environments and analyze agent behavior, but they never list which environments belong to which group. A reader cannot verify the claims about “synthesis tasks” without guessing or cross-referencing external knowledge of the specific environments (CarRacing, MiniGrid, etc.). This split is central to the paper’s diagnostic argument, so its definition must be explicit.

Additionally, the operational definitions of “synthesis edit” and “parametric edit” rely on the concept of “stripped AST topology.” While the text mentions that “numeric constants are stripped,” it does not specify the exact procedure (e.g., are function names stripped? are control flow structures preserved?). This lack of precision makes the quantitative results in Table 4 difficult to interpret or reproduce.

Finally, the term “gait” in the BipedalWalker case study (Section 3.2) is used as a critical milestone (“viable gait topology”) without a formal definition. In robotics, a gait is a specific pattern of movement, but the paper treats it as a binary state determined by a return threshold. Defining this threshold and the pattern it represents would make the case study self-contained.

The notation in Section 4.1 is standard but slightly dense; a brief clarification of the mapping Φ would improve clarity for non-specialists. Overall, these are minor omissions that can be fixed with a few sentences of definition.

Action items.

- **[writing]** Section 3.1 (Analysis) introduces ‘synthesis-dominant’ and ‘tuning-dominant’ groups without defining the criteria for this split. An adjacent-field reader cannot determine which of the 16 environments fall into which category or why. Add a sentence listing the environments in each group or refer to a table defining this split.
- **[writing]** Section 3.1 defines ‘synthesis edit’ and ‘parametric edit’ based on ‘stripped AST topology’ but does not define the stripping procedure (e.g., ‘numeric constants are stripped’ is mentioned later but the specific algorithm or regex logic is opaque). Clarify what constitutes a ‘stripped topology’ or provide a brief example of the transformation.
- **[writing]** Section 3.2 uses the term ‘gait-producing topology’ and ‘viable gait’ without defining what constitutes a ‘gait’ in the context of BipedalWalker or how the ‘return threshold’ for viability is determined. Define ‘gait’ operationally (e.g., ‘a periodic oscillation in joint angles resulting in positive forward velocity’) and state the specific return value used as the threshold.
- **[writing]** Section 4.1 (Framework) introduces the notation $R_i = hi(W_i)$ and π_{heta} without explicitly defining the domain and codomain of the function hi or the parameter space of π_{heta} . While context implies these, a formal definition (e.g., ‘ hi maps workspace states to executable policy systems’) would prevent ambiguity for readers from adjacent fields.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s argument structure is largely coherent, with premises and conclusions generally following logically. However, several sections contain minor logical gaps or ambiguous phrasing that could mislead readers about the data’s implications.

In Section 4.1, the claim that “Nontrivial code volume is therefore not sufficient for strong performance” is drawn from Table 3, which shows GPT-5.5 (high volume) outperforming others. The text implies a negative correlation between volume and success, but the data shows a positive correlation for the top performer. This creates a non-sequitur: the premise (high volume correlates with high performance) does not support the conclusion (volume is not sufficient). The authors likely intend to say that volume alone is insufficient without correct structure, but the current phrasing contradicts the observed trend. Rephrase to clarify that while volume is necessary for synthesis tasks, it does not guarantee success without appropriate structural design.

Section 4.2 states “GPT-5.5 is the only run... with a positive high-return gait” for Bipedal-Walker. Table 1 confirms GPT-5.5’s score (248.874) is positive while others are negative, so the claim is factually correct. However, the phrasing “only run” is ambiguous without explicitly stating the negative scores of competitors in the same sentence, which could lead readers to overlook the contrast. Adding a brief reference to the negative scores of other agents would strengthen the logical connection.

In Section 4.3, the text claims MiniMax-M3 and DeepSeek-V4-Pro “mostly churn structure without traction (10% and 3%)” for synthesis edits. Table 4 shows hit rates of 10% and 3%, which are indeed low, but “mostly churn” is a subjective interpretation. The logical gap is that 10% success rate does not necessarily mean “mostly churn” if the remaining 90% are neutral or slightly negative. Clarify the threshold for “churn” or rephrase to “low hit rates” to avoid overstatement.

Section 5.1’s claim about GPT-5.5’s “nine wins and top-two placement on all 16 environments” is supported by Table 2, but the term “wins” could be ambiguous without explicit definition as “first-place finishes.” While the table clarifies this, the text should define “wins” to prevent misinterpretation, especially for readers unfamiliar with the leaderboard metrics.

Overall, the paper’s reasoning holds together with minor adjustments needed to clarify ambiguous phrasing and ensure conclusions align precisely with the data presented. No fatal logical breaks are present, but these revisions would enhance the argument’s clarity and rigor.

Action items.

- **[writing]** The paper’s argument structure is largely coherent, with premises and conclusions generally following logically. However, several sections contain minor logical gaps or ambiguous phrasing that could mislead readers about the data’s implications. In Section 4.1, the claim that “Nontrivial code volume is therefore not sufficient for strong performance” is drawn from Table 3, which shows GPT-5.5 (high volume) outperforming others. The text implies a negative correlation between volume and success,

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper’s rhetoric frequently exceeds the scope of the evidence provided in its own tables, particularly regarding the performance of GPT-5.5. The abstract and introduction repeatedly assert that GPT-5.5 achieved “top-two performance on all 16 environments.” However, a direct inspection of Table 2 (Core16 Raw) reveals that GPT-5.5 scored 9.818 on “Roundabout” (third place behind DeepSeek-V4-Pro’s 10.203) and -37.106 on “Pusher” (third place behind Claude Opus 4.7’s -38.520). This is a clear case of scope overreach where a specific, limited dataset result is framed as a universal dominance. The claim must be corrected to reflect the actual count (14/16) or qualified to exclude the specific environments where it failed to place top-two.

Additionally, the conclusion generalizes the mechanism of “preserving useful candidates under budget pressure” as a universal trait of strong agents. The evidence for this is derived exclusively from the 128-episode budget constraint of the Core16 suite. The paper does not test different budget sizes or interaction limits, yet the conclusion implies this behavior is intrinsic to the agents’ capability rather than a result of the specific experimental constraint. The text should be hedged to specify that this observation holds “under the fixed 128-episode budget” to avoid implying a general law of agent behavior.

Finally, while the paper correctly identifies that MiniMax-M3 and DeepSeek-V4-Pro each secured one “win” (first place), the narrative framing in Section 4.1 (“each win one environment”) risks obscuring the fact that MiniMax-M3 also achieved top-two placements in other environments (Parking, FetchPickAndPlace). While not a fatal error, the phrasing creates a binary “win/loss” narrative that slightly over-simplifies the nuanced ranking data presented in the tables. Precision in describing partial success is necessary to match the granularity of the reported metrics.

Action items.

- **[writing]** Abstract and Intro claim GPT-5.5 has ‘top-two performance on all 16 environments,’ but Table 2 shows it is 3rd on Roundabout and Pusher. Correct the claim to ‘14 of 16’ or remove the universal quantifier.
- **[writing]** Conclusion states strong agents ‘preserve useful candidates under budget pressure’ as a general rule. Evidence is limited to a fixed 128-episode budget. Scope the claim to ‘under the 128-episode budget tested’ to avoid overgeneralization.
- **[writing]** Section 4.1 says agents ‘each win one environment,’ which oversimplifies MiniMax-M3’s top-2 finishes in Parking and FetchPickAndPlace. Clarify to ‘each secure exactly one first-place win’ to match the table’s nuance.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a benchmark (EvoPolicyGym) for evaluating autonomous agents that iteratively edit executable policies in simulated reinforcement learning environments. The work is fundamentally a methodological and evaluation study using standard, open-source environments (Gymnasium, MuJoCo, MiniGrid) and synthetic interaction loops.

From a safety and ethics perspective, the research does not present foreseeable, non-trivial risks of harm that are unaddressed. The “policies” being evolved are decision-making scripts for simulated agents (e.g., driving a car in a video game, navigating a grid world, or controlling a robotic arm in a physics simulator). These are not real-world control systems, biological agents, or tools for generating disinformation, malware, or surveillance capabilities. The “dual-use” potential of a benchmark for code-editing agents is generic to the field of LLM evaluation and

does not constitute a specific, actionable risk requiring unique mitigation in this context.

The paper uses no human subjects, no personal data, and no sensitive datasets. The data generated (trajectories, code edits) is synthetic and derived from the interaction of agents with the provided environments. There are no issues regarding consent, IRB approval, or data privacy. The code and data artifacts are linked to public repositories, and the paper does not appear to violate any licensing terms regarding the use of the underlying environment libraries (which are standard in the field).

While the paper evaluates models that may have broader capabilities (e.g., GPT-5.5, Claude Opus 4.7), the scope of this specific work is limited to their performance on a controlled, simulated benchmark. The paper does not provide operational details for exploiting vulnerabilities, nor does it demonstrate the creation of harmful capabilities. The “limitations” section appropriately notes the diagnostic nature of the analysis and the scope of the environments.

Consequently, there are no specific safety or ethics disclosures missing, nor are there any unmitigated risks identified that would require revision. The work falls squarely within the category of low-risk, standard ML benchmarking research.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The central claim that EvoPolicyGym effectively evaluates “autonomous policy evolution” and that specific agents demonstrate superior “structural synthesis” capabilities is currently unsupported by the experimental design presented. The primary evidentiary gap is the complete absence of variance reporting. Section 4.1 and Table 1 present leaderboard results derived from exactly one run per agent-environment pair ($n=1$). In stochastic environments and with non-deterministic LLM agents, a single run is insufficient to establish a general capability. The reported “strongest aggregate rank” for GPT-5.5 could easily be an artifact of a favorable random seed, a lucky initial prompt, or a specific trajectory that did not generalize. Without reporting results across multiple seeds (e.g., 3-5) with mean and standard deviation, the reader cannot distinguish a robust effect from noise.

Furthermore, the analysis in Section 4.2 attempts to attribute performance differences to the mechanism of “structural synthesis” versus “parametric tuning.” However, the experimental design introduces a severe confound: the top-performing agents (GPT-5.5) use the Codex harness, while the others use the Claude Code harness. Additionally, Appendix Table 3 shows massive disparities in token usage (e.g., Claude Opus using ~28M cache tokens vs. GPT-5.5 using ~11M). The observed correlation between “synthesis edits” and high scores may simply reflect the superior tool-use capabilities or context management of the Codex harness, or the sheer volume of tokens processed, rather than the specific “synthesis” mechanism the authors claim to isolate. To support the claim that the *mechanism* of synthesis is the driver, the authors must control for

the harness and token budget, or run ablations where the same model uses different harnesses.

Finally, the classification of environments into “synthesis-dominant” and “tuning-dominant” groups (Section 4.2) lacks an a priori definition. The text implies these categories are based on task requirements (e.g., “pixel-perception and symbolic-planning”), but the grouping aligns perfectly with where the top agents succeeded. If the categories were defined post-hoc based on the results, the subsequent analysis showing a performance gap is circular. The authors must define these task categories based on intrinsic environment properties (e.g., observation space dimension, need for long-term memory) independent of the agent outcomes to validate the diagnostic utility of their split.

Action items.

- **[science]** The central claim that EvoPolicyGym effectively evaluates “autonomous policy evolution” and that specific agents demonstrate superior “structural synthesis” capabilities is currently unsupported by the experimental design presented. The primary evidentiary gap is the complete absence of variance reporting. Section 4.1 and Table 1 present leaderboard results derived from exactly one run per agent-environment pair ($n=1$). In stochastic environments and with non-deterministic LLM agents, a single ru

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical treatment of the experimental results is insufficient to support the definitive performance claims. The primary issue is the complete absence of variance reporting for the main quantitative results.

In Section 4.2 and Tables 1-2, the authors report precise held-out returns (e.g., “248.874”) derived from a single 128-episode run per agent-environment pair. Given the stochastic nature of LLM agents and RL environments, a single run cannot distinguish signal from noise. Reporting a single point estimate without standard deviation, standard error, or confidence intervals implies a false precision. Standard practice requires reporting mean $\pm$ SD over multiple seeds (typically 3-5) to establish robustness.

Additionally, Table 3 presents “hit rates” (e.g., 41% for GPT-5.5 synthesis edits) as fixed metrics derived from a single run’s trajectory. Without binomial confidence intervals, readers cannot assess if observed differences (e.g., 41% vs 10%) are statistically significant or within sampling noise.

Finally, the normalization formula in Section 4.1 uses R_e^{best} (the maximum score achieved by any evaluated agent) as the denominator. This introduces a circular dependency where the scale is determined by the sample maximum, artificially compressing scores and distorting relative performance gaps. The normalization should use a fixed, external upper bound to ensure independence from the specific sample of agents tested.

To resolve these issues, the authors must re-run experiments with multiple seeds to generate variance estimates, recalculate edit-type hit rates with confidence intervals, and revise the normalization protocol.

Action items.

- **[science]** Section 4.2 reports single point estimates from one run per cell. Report mean $\pm$ SD over 3 seeds for all leaderboard metrics to distinguish signal from noise, or explicitly state results are illustrative single-seed runs.
- **[writing]** Table 3 reports edit ‘hit rates’ (e.g., 41%) from single runs without uncertainty. Add binomial confidence intervals (e.g., Wilson score) to these rates to assess if differences between agents are statistically distinguishable.
- **[science]** The normalization in Section 4.1 uses the sample maximum (R_e^{best}) as the denominator, creating a circular dependency. Replace with a fixed external anchor (e.g., human expert score) to avoid sample-dependent scaling artifacts.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-structured and readable, with a clear logical flow from the problem definition to the experimental results and analysis. The abstract effectively summarizes the contribution, and the section transitions are mostly smooth. However, there are several instances where sentence construction impedes immediate comprehension, and a few unremoved author comments remain in the text.

In Section 3.1, the paragraph discussing code structure contains redundancy. The final two sentences (“Nontrivial code volume is therefore not sufficient... In other words, complex code is not necessarily...”) restate the same point without adding new information. This forces the reader to process the same idea twice. Merging these into a single, concise statement would improve the paragraph’s efficiency.

Similarly, in Section 3.2, the explanation of the case study selection is overly verbose and passive (“The two timelines are randomly sampled case studies rather than an additional selection based on...”). A more direct active voice construction would clarify the authors’ intent and reduce cognitive load.

In Section 4.1, a long run-on sentence attempts to explain the decision not to normalize token usage. Splitting this into two sentences—one stating the decision and the other explaining the rationale—would make the argument easier to parse on the first pass.

Finally, there are remnants of author comments (e.g., in Section 2 regarding the term “coding agent”) that should be removed before publication. While these do not obscure the meaning entirely, they break the professional flow of the prose. Addressing these specific points will ensure

the paper moves through the reader's mind without friction.

Action items.

- **[writing]** The paper is generally well-structured and readable, with a clear logical flow from the problem definition to the experimental results and analysis. The abstract effectively summarizes the contribution, and the section transitions are mostly smooth. However, there are several instances where sentence construction impedes immediate comprehension, and a few unremoved author comments remain in the text. In Section 3.1, the paragraph discussing code structure contains redundancy. The final two sente